

Compact RF Admittance level switch

For Solid Application Model: RFLS-100S

PRODUCT DESCRIPTION:



The RFLS-100S RF Admittance type level switch is a simple, low cost, rugged point Level Switch designed for Level detection of Bulk materials in a bins or silo. This level switch can be used for detecting high, low level or for differential level detection and is used in various level detection applications like. cement industries, material handling plants, chemical industry, power utilities industry, food industry, pulp/paper industry, pharmaceuticals industry, water/ treatment plants, foundry industry etc.

The specialty of the probe is that it tends to ignore the build-up on the probe, thus making it suitable for application, which tends to stick on the probe.

The unit has integral electronics along with time delay feature, LED indication, fail safe mode **etc.**

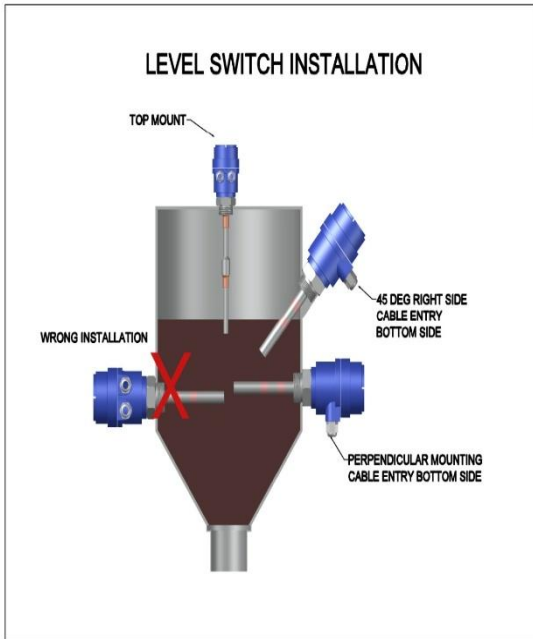
SALIENT FEATURES:

- Advanced RF Admittance Technology
- Compact and Heavy-Duty Design for Solid Applications
- Wide Range for Industrial Applications
- Easy Installation and Maintenance
- Weatherproof / Flameproof Enclosure as per Requirement
- PTFE-Insulated Electrode for High-Temperature Applications up to 180°C
- No Moving Parts – No Mechanical Wear and Tear
- Adjustable Differential Time Delay for Output
- Field-Configurable Fail-Safe High / Low Selection
- Reliable Operation Even with Sticky and Conductive Material Build-Up



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PROBE CONSTRUCTION SUITABLE FOR MATERIAL BUILT UP:



The RFLS-100S Normal probe tends to fail when there is a build-up on the probe. The COAT Guard electrode senses the material build up on the probe by employing a special Build up sensing technology. The COAT guard unit is insulated from the active probe and the ground is driven with same RF frequency as that of active electrode and having the same polarity. When a material build up is formed on the probe, the RF current from the COAT guard unit tends to saturate the build-up area between COAT guard probe and silo wall, thereby no electrical energy will flow from active probe to tank wall. When the material level goes up and touches the sense probe a RF current will flow from the sense probe to ground over and above the saturated region formed by the Built-up sensing circuitry.

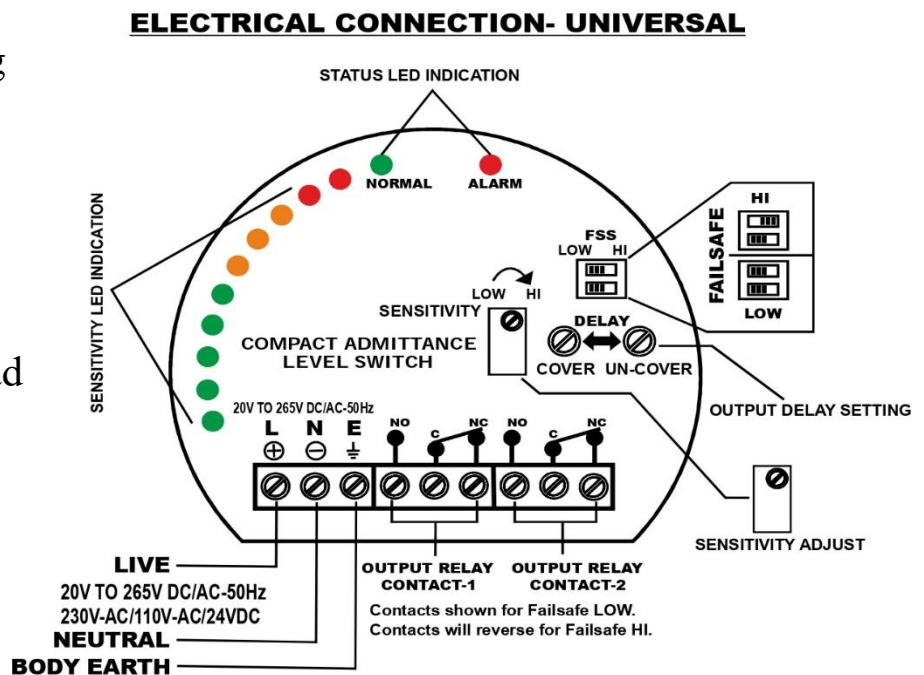
THE PART OF COMPACT LEVEL SWITCH ARE EXPLAINED IN FIGURE



- ---Aluminum Housing
- -----Cable Entry
- -----Mounting Thread
- -----PTFE insulation
- ---Shield
- -----PTFE insulation
- -----sense

ELECTRICAL CONNECTION

Universal /230 VAC/110VAC/24 VDC



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OPERATING PRINCIPLE

The RFLS-100S RF Admittance Level Switch operates on the RF Admittance Sensing Principle for reliable point-level detection of solid materials, such as powders, granules, pellets and bulk solids. A high-frequency RF signal is applied to the sensing probe. When the probe is surrounded by air, the sensing circuit remains in its normal reference condition. As the solid material reaches the sensing probe, the electrical admittance between the probe and vessel ground changes according to the material properties. The electronic circuit continuously detects this change. When the admittance change reaches the preset Sensitivity Threshold, the electronic circuit detects the presence of material and operates the Output Relay, providing a level signal for high-level or low-level control. The Active Shield/Guard minimizes the effect of material coating or buildup on the sensing probe, reducing false switching and providing stable, reliable and repeatable level detection

The RFLS-100S provides adjustable sensitivity and Fail-Safe High/Low selection to suit different solid material and process requirements.

DATA SHEET

RFLS-100 – RF Admittance Level Switch

Selection / Ordering Information ()

Selection	Code	Description
Model No.	RFLS-100	RF Admittance Level Switch
Application	S	Solids
Mounting Type	IN	Integral
Probe Type	RD / RP/HD	Rigid Rod / Rope Flexible/Heavy Duty type Probe
Insulation	FP/PT/ CE	Full PTFE / Part PTFE / Ceramic
Probe Diameter	D1 / D2	Standard 16mm / Heavy Duty 30mm
Probe Material	04/ 16/ 6L/MS/ SM	SS304/ SS316/ SS316L /Mild steel/ Special Material

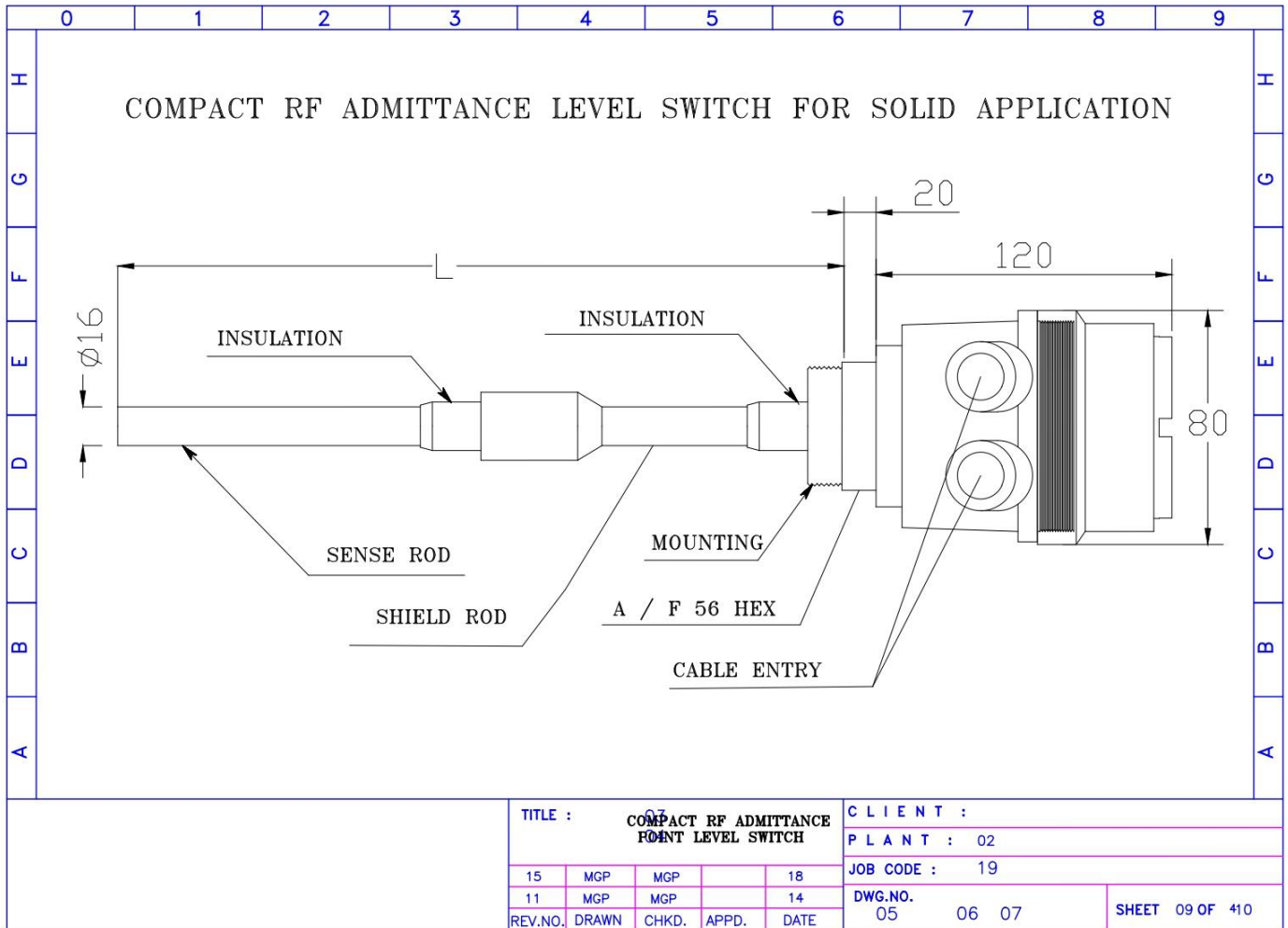
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Selection	Code	Description
Power Supply	AC/DC/A1/UN	230VAC / 24VDC / 110VAC / Universal
Output	DP/SP/PN	DPDT Relay /SPDT Relay / PNP Relay
Switching Delay	ND / DA	No Delay / Adjustable
Indication	L	LED
Enclosure	WP / FL	Weatherproof / Flameproof
Connection	T	Threaded
	F	Flanged
Mounting	B1 / B2 / B3/B4	1" / 1½" / 2" BSP/ Special Mounting
	N1 / N2 / N3/N4	1" / 1½" / 2" NPT/ Special Mounting
Flange Rating	A1 / A3 / A6	ANSI 150# / ANSI 300# /ANSI 600#
Flange Facing	RF / FF	Raised Face / Flat Face
Flange Material	S4 / S6/ S16/MS/SL	SS304 / SS316 /SS316L /Mild steel / Special material
Flange Size	01 / 15 / 02 / 03 / 04/PS	1" / 1½" / 2" / 3" / 4"/ Special Process
Pressure	P1 /P2/P3/P4	Standard 0 to 2KG/ 2 To 10kg /10kg to 40kg / Customised
Temperature	T1/ T2/TS	Max -20–150°C / 200°C/ Customer Specified Special Designed
High Temperature stand off	H8/HS	80MM/Special standoff Extension
Cable Gland	G1 / G2 / G3	PG13.5 / ½"NPT / ¾" ET
Ground / Reference	GS / GX	Standard / Special
Material Certificate	M0 / MC	Not Required / Material test certificate
Test Certificate	TC / CC	Test Certificate / Calibration Certificate
Documentation	DS / GA / IR	Datasheet / GA / Inspection Report
Tagging	TAG	Customer Tag No.
Probe Length	XXXX	200mm To Customize

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Mechanical Drawing:



Manufactured By:

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